

Week 7: Effect Modification

Davood Tofighi, Ph.D.

R Packages

```
# Load required packages using pacman
pacman::p_load(
  dagitty, # DAG construction and analysis
  ggdag, # Beautiful DAG plotting with ggplot2
  tidyverse, # Data manipulation and visualization
  broom, # Tidy model outputs
  grf, # Generalized random forests for causal ML
  rpart, # Decision trees for causal trees
  rpart.plot, # Plotting decision trees
  kableExtra, # Enhanced tables
  gt, # Grammar of tables
  patchwork, # Combining plots
  marginaleffects # Marginal effects and contrasts
)
```

Learning Objectives

By the end of this week, students will be able to:

1. **Define** effect modification and distinguish it from confounding using formal notation and potential outcomes framework
2. **Compare** additive versus multiplicative scales of interaction and explain when each is most appropriate

3. **Differentiate** between statistical interaction and mechanistic interaction using Rothman’s sufficient cause framework
4. **Calculate** measures of effect modification by hand using simple 2×2 tables and potential outcomes
5. **Implement** R code to detect, visualize, and quantify effect modification in observational data
6. **Apply** modern causal machine learning methods (causal trees, causal forests) to discover heterogeneous treatment effects
7. **Interpret** interaction terms in regression models within a causal framework

1 Comprehensive Topic Explanation

1.1 What is Effect Modification?

Effect modification, also known as **treatment effect heterogeneity**, occurs when the causal effect of an exposure varies across different levels or values of another variable, called the **effect modifier** (Hernán & Robins, 2025, Chapter 4). Unlike confounding, which creates bias that must be controlled, effect modification represents genuine biological or social phenomena where treatment effects differ across subgroups.

Formally, let V be a potential effect modifier. We say that V modifies the effect of treatment A on outcome Y if:

$$\mathbb{E}[Y^{a=1} - Y^{a=0} \mid V = v] \neq \mathbb{E}[Y^{a=1} - Y^{a=0} \mid V = v']$$

for some values $v \neq v'$ of the modifier V .

This definition is **inherently causal** because it compares potential outcomes within strata of the modifier. Effect modification cannot be definitively established from purely observational associations—it requires the same causal assumptions (exchangeability, positivity, consistency) that we need for identifying any causal effect.

1.2 Scales of Interaction: Additive vs. Multiplicative

The same data can show interaction on one scale but not another, leading to the crucial question: **which scale matters?**

Additive Scale: Interaction exists if the joint effect differs from the sum of individual effects:

$$\mathbb{E}[Y^{a=1,v=1}] - \mathbb{E}[Y^{a=0,v=0}] \neq (\mathbb{E}[Y^{a=1,v=0}] - \mathbb{E}[Y^{a=0,v=0}]) + (\mathbb{E}[Y^{a=0,v=1}] - \mathbb{E}[Y^{a=0,v=0}])$$

Multiplicative Scale: Interaction exists if the joint effect differs from the product of individual relative effects:

$$\frac{\mathbb{E}[Y^{a=1,v=1}]}{\mathbb{E}[Y^{a=0,v=0}]} \neq \frac{\mathbb{E}[Y^{a=1,v=0}]}{\mathbb{E}[Y^{a=0,v=0}]} \times \frac{\mathbb{E}[Y^{a=0,v=1}]}{\mathbb{E}[Y^{a=0,v=0}]}$$

For **rare outcomes**, the additive scale is often more relevant for public health because it directly quantifies excess cases attributable to interaction. For **common outcomes** or when thinking about biological mechanisms, multiplicative interactions may be more natural.

1.3 Statistical vs. Mechanistic Interaction

Statistical interaction refers to departure from additivity (or multiplicativity) in a fitted model. It's what we observe in regression output when interaction terms are significant.

Mechanistic interaction refers to biological or causal pathways where factors truly work together to produce effects. Rothman's sufficient cause framework provides the clearest illustration of mechanistic interaction.

1.4 Rothman's Sufficient Cause Framework

Kenneth Rothman introduced "causal pies" to represent sufficient causes—combinations of factors that, together, inevitably produce

the outcome (VanderWeele, 2015, Chapter 1). Each “pie” represents one pathway to disease, with each “slice” representing a component cause.

Synergism occurs when two factors appear as component causes in the same causal pie. For example:

- Pie 1: {Smoking, Asbestos exposure, Genetic susceptibility} → Lung cancer
- Pie 2: {Heavy alcohol use, Hepatitis B, Poor nutrition} → Liver disease

If smoking and asbestos appear together in Pie 1 but neither appears in other pies, they exhibit synergistic interaction.

This framework explains why:

- Interaction can exist on the additive scale even when multiplicative interaction is absent
- Mechanistic interaction may not always manifest as statistical interaction
- The “background” prevalence of other component causes affects observed interaction patterns

Visualizing Rothman’s Sufficient Cause Model

Explanation Of the Graphic

This graphic perfectly illustrates the key concepts:

1. **Multiple Sufficient Causes:** There are three distinct causal pathways (Pie I, II, and III) that can lead to the outcome.
2. **Component Causes:** Each slice (A, B, C, etc.) is a component cause. Notice that Pie III has four component causes, while the others have three, showing that the number of components required to complete a mechanism can vary.
3. **Interaction:** In Pie I, component causes A, B, and C interact. If an individual has A and C, the addition of B completes the pie and causes the disease. This is a mechanistic interaction.

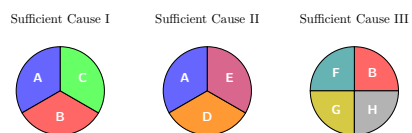


Figure 1: A visual representation of Rothman's sufficient-component cause model. Each pie is a distinct sufficient cause. Each slice is a component cause. Note that a component cause (e.g., A or B) can be part of multiple sufficient causes.

4. **No Necessary Cause:** In this specific example, there is no single component cause that appears in *all three* pies, so there is no necessary cause shown. If “A” were also a slice in Pie III, it would be a necessary cause for this disease.

This type of visualization is incredibly useful for teaching, for clarifying your own thinking about a research problem, and for communicating complex causal ideas to a broader audience.

1.5 Causal Interaction with Multiple Treatments

When we have two treatments A and B , we can define four potential outcomes for each individual:

- $Y^{a=0,b=0}$: Neither treatment
- $Y^{a=1,b=0}$: Only treatment A
- $Y^{a=0,b=1}$: Only treatment B
- $Y^{a=1,b=1}$: Both treatments

Additive causal interaction exists when:

$$\mathbb{E}[Y^{a=1,b=1} - Y^{a=0,b=0}] \neq (\mathbb{E}[Y^{a=1,b=0} - Y^{a=0,b=0}] + \mathbb{E}[Y^{a=0,b=1} - Y^{a=0,b=0}])$$

The left side represents the **joint effect**, while the right side represents the **sum of marginal effects**.

1.6 Modern Causal Machine Learning Approaches

Traditional approaches to effect modification require **a priori** specification of interaction terms. Modern causal machine learning methods can **discover** heterogeneous treatment effects without pre-specifying interactions.

Causal Trees: Recursive partitioning that splits on covariates to maximize treatment effect heterogeneity rather than outcome prediction (Athey & Imbens, 2016).

Causal Forests: An ensemble method that averages many causal trees, providing both point estimates and uncertainty quantification for heterogeneous treatment effects (Wager & Athey, 2018).

These methods are particularly valuable when:

- The modifier space is high-dimensional
- Interactions involve multiple variables simultaneously
- The functional form of modification is unknown

2 Intuitive Clarifications of Challenging Ideas

💡 □ Learning Tip: The Interaction Paradox

Analogy: Think of effect modification like cooking. Salt enhances the flavor of tomatoes more than it enhances the flavor of ice cream. The “effect” of salt depends on what you’re adding it to—this is effect modification.

However, whether we measure this on a “taste intensity” scale (additive) or a “flavor multiplication” scale (multiplicative) can lead to different conclusions about whether interaction exists.

2.1 Why Confounding Effect Modification

Common Misconception: “If a variable is associated with both treatment and outcome, it’s a confounder.”

Reality: Effect modifiers can be associated with treatment and outcome WITHOUT being confounders. The key difference:

- **Confounder:** Creates bias in the marginal treatment effect
- **Effect Modifier:** Changes the treatment effect across strata, but each stratum-specific effect may be unbiased

Visual Metaphor: Think of confounding as wearing tinted glasses that distort ALL colors equally. Effect modification is like having

different lighting in different rooms—the same object looks different in each room, but you’re seeing it clearly in each context.

2.2 The Scale Dependence Problem

Consider these data:

	Risk with Group Treatment	Risk without Treatment	Risk Difference	Risk Ratio
Men	0.20	0.10	0.10	2.0
Women	0.15	0.05	0.10	3.0

- **Additive scale:** No interaction (both differences = 0.10)
- **Multiplicative scale:** Strong interaction (ratios = 2.0 vs 3.0)

Intuition: This happens because the baseline risks differ between men and women. When baseline risk is lower, the same absolute increase represents a larger relative increase.

2.3 Why Sufficient Causes Matter

Mechanical Analogy: Think of disease as a car failing to start. There might be multiple ways for the car to fail:

- Path 1: {Dead battery} + {Cold weather} + {Old alternator}
- Path 2: {Empty gas tank} + {Clogged fuel filter}
- Path 3: {Faulty starter} + {Corroded connections}

If “cold weather” and “dead battery” appear together in Path 1, they’re synergistic—cold weather only matters when you also have a dead battery problem.

This explains why some interactions only appear in certain populations or contexts.

3 Simple, Hand-Calculation Numerical Example

Let's work through a complete example calculating effect modification by hand.

Worked Example: Drug Efficacy by Age Group

Scenario: A new antihypertensive drug is tested. We want to know if age modifies its effect.

Data:

Age Group	Treatment	n	Mean BP Reduction
Young (<50)	Drug	100	15 mmHg
Young (<50)	Placebo	100	3 mmHg
Old (≥ 50)	Drug	100	20 mmHg
Old (≥ 50)	Placebo	100	8 mmHg

Step 1: Calculate stratum-specific effects

Young group effect: $15 - 3 = 12$ mmHg reduction

Old group effect: $20 - 8 = 12$ mmHg reduction

Step 2: Test for effect modification

Since both effects equal 12 mmHg, there is **no effect modification on the additive scale**.

Step 3: Check multiplicative scale

Young group ratio: $15/3 = 5.0$

Old group ratio: $20/8 = 2.5$

Since $5.0 \neq 2.5$, there is **effect modification on the multiplicative scale**.

Interpretation: The drug provides the same absolute benefit (12 mmHg) regardless of age, but represents a larger relative improvement in younger patients due to their lower baseline response to placebo.

Hand Calculation: Sufficient Cause Framework

Scenario: Consider lung cancer with two exposures: smoking (S) and asbestos (A).

Assume three sufficient cause “pies”:

- Pie 1: {S, A, U_1 } (smoking + asbestos + unknown factors)
- Pie 2: {S, U_2 , U_3 } (smoking + other unknown factors)
- Pie 3: { U_4 , U_5 , U_6 } (non-smoking causes)

Population prevalences:

- S: 30%, A: 10%, U_1 : 5%, U_2 : 2%, U_3 : 40%, U_4 : 1%, U_5 : 20%, U_6 : 30%

Step 1: Calculate disease probabilities

$$\begin{aligned}
 \mathbb{P}(\text{Disease} \mid S = 1, A = 1) &= \mathbb{P}(\text{Pie 1} \vee \text{Pie 2}) \\
 &= \mathbb{P}(\text{Pie 1}) + \mathbb{P}(\text{Pie 2}) - \mathbb{P}(\text{Pie 1} \wedge \text{Pie 2}) \\
 &= \mathbb{P}(U_1 = 1) + \mathbb{P}(U_2 = 1 \wedge U_3 = 1) - \mathbb{P}(U_1 = 1 \wedge U_2 = 1 \wedge U_3 = 1) \\
 &= 0.05 + (0.02 \times 0.40) - (0.05 \times 0.02 \times 0.40) \\
 &= 0.05 + 0.008 - 0.0004 \\
 &= 0.0576
 \end{aligned}
 \tag{1}$$

$$\mathbb{P}(\text{Disease} \mid S = 1, A = 0) = \mathbb{P}(\text{Pie 2}) = \mathbb{P}(U_2 = 1 \wedge U_3 = 1) = 0.02 \times 0.40 = 0.008$$

$$\mathbb{P}(\text{Disease} \mid S = 0, A = 1) = \mathbb{P}(\text{Pie 3}) = \mathbb{P}(U_4 = 1 \wedge U_5 = 1 \wedge U_6 = 1) = 0.01 \times 0.20 \times 0.30 = 0.0006$$

$$\mathbb{P}(\text{Disease} \mid S = 0, A = 0) = \mathbb{P}(\text{Pie 3}) = 0.0006$$

Step 2: Calculate interaction measures

Joint effect: $0.0576 - 0.0006 = 0.057$

Smoking alone: $0.008 - 0.0006 = 0.0074$

Asbestos alone: $0.0006 - 0.0006 = 0$

Sum of individual effects: $0.0074 + 0 = 0.0074$

Synergy index: $(0.057 - 0.0074)/0.0074 = 6.7$

This indicates strong synergistic interaction—the combined effect is nearly 7 times larger than expected from adding individual effects.

4 R Code Examples & Interpretation

4.1 Simulating Effect Modification Data

```
# Set seed for reproducibility
set.seed(12345)

# Simulate data with age-dependent treatment effect
n <- 1000
age <- rnorm(n, mean = 50, sd = 15)
age_centered <- age - 50

# Treatment assignment (slightly more likely for older
patients)
treatment <- rbinom(n, 1, prob = 0.4 + 0.001 *
age_centered)

# Outcome with effect modification
# Base outcome depends on age
# Treatment effect is stronger in older patients
baseline_outcome <- 120 + 0.5 * age_centered + rnorm(n,
0, 10)
```

```

treatment_effect <- -10 - 0.2 * age_centered # Stronger
effect in older patients
outcome <- baseline_outcome + treatment *
treatment_effect

# Create dataset
sim_data <- tibble(
  id = 1:n,
  age = age,
  age_group = ifelse(age ≥ 50, "Older", "Younger"),
  treatment = factor(treatment, labels = c("Control",
  "Treatment")),
  outcome = outcome
)

# Show first few rows
# head(sim_data) >
# kable(digits = 2, caption = "Simulated Dataset
with Effect Modification") >
# kable_styling(bootstrap_options = "striped")

head(sim_data)

```

Table 3: Simulated Dataset with Effect Modification

id	age	age_group	treatment	outcome
1	58.78293	Older	Treatment	110.88865
2	60.64199	Older	Treatment	106.48643
3	48.36045	Younger	Control	124.25448
4	43.19754	Younger	Treatment	120.43361
5	59.08831	Older	Control	112.06140
6	22.73066	Younger	Treatment	82.47201

4.2 Exploratory Analysis of Effect Modification

```
# Visualize potential effect modification
p1 <- sim_data %>%
  ggplot(aes(x = treatment, y = outcome, fill =
    treatment)) +
  geom_boxplot(alpha = 0.7) +
  facet_wrap(~age_group) +
  labs(
    title = "Treatment Effects by Age Group",
    y = "Blood Pressure (mmHg)",
    x = "Treatment Group"
  ) +
  theme_minimal() +
  scale_fill_brewer(type = "qual", palette = "Set2") +
  theme(legend.position = "none")

# Scatter plot with regression lines
p2 <- sim_data %>%
  ggplot(aes(x = age, y = outcome, color = treatment)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Treatment Effect Varies with Age",
    x = "Age (years)",
    y = "Blood Pressure (mmHg)",
    color = "Treatment"
  ) +
  theme_minimal() +
  scale_color_brewer(type = "qual", palette = "Set1")

# Combine plots
p1 / p2
```

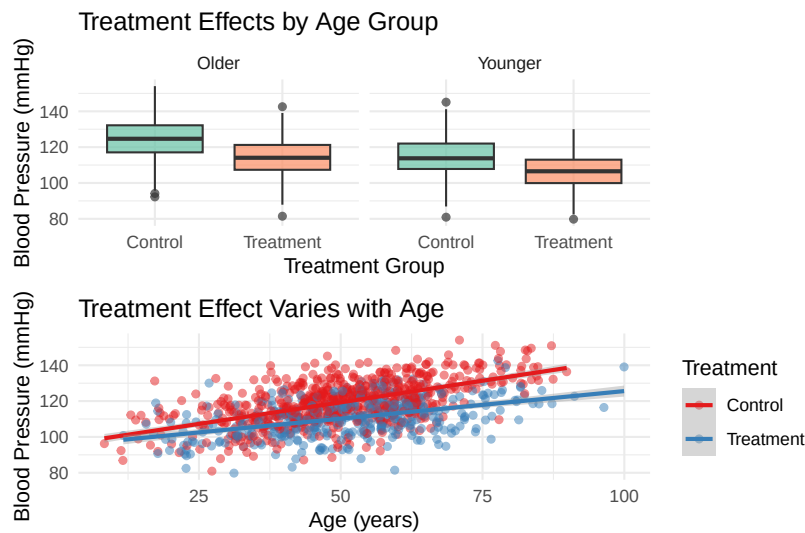


Figure 2: Exploring Effect Modification Patterns

This visualization immediately suggests that the treatment effect differs between age groups. The older group shows a larger treatment effect (greater separation between treatment and control).

4.3 Statistical Testing for Effect Modification

```
# Model without interaction (assumes no effect
modification)
model_no_int <- lm(outcome ~ treatment + age, data =
sim_data)

# Model with interaction (tests for effect modification)
model_with_int <- lm(outcome ~ treatment * age, data =
sim_data)

# Compare models
# anova(model_no_int, model_with_int) >
#   tidy() >
#   kable(digits = 4, caption = "ANOVA Comparing
Models") >
```

```
# kable_styling(bootstrap_options = "striped")

anova(model_no_int, model_with_int)
# Show interaction model results
# tidy(model_with_int) ▷
# kable(digits = 4, caption = "Interaction Model
Coefficients") ▷
# kable_styling(bootstrap_options = "striped")
summary(model_with_int)
```

Call:

```
lm(formula = outcome ~ treatment * age, data = sim_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-31.697	-6.449	0.081	6.732	27.702

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	95.22094	1.42254	66.94	< 2e-16 ***
treatmentTreatment	-0.40111	2.23151	-0.18	0.857
age	0.48182	0.02714	17.75	< 2e-16 ***
treatmentTreatment:age	-0.17408	0.04205	-4.14	3.76e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.811 on 996 degrees of freedom

Multiple R-squared: 0.3766, Adjusted R-squared: 0.3747

F-statistic: 200.6 on 3 and 996 DF, p-value: < 2.2e-16

Table 4: Regression Models Testing Effect Modification

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
997	97514.60	NA	NA	NA	NA

Table 4: Regression Models Testing Effect Modification

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
996	95864.72	1	1649.879	17.14165	3.76e-05

Interpretation: The significant interaction term (treatmentTreatment:age) indicates that age modifies the treatment effect. For each additional year of age, the treatment effect becomes 0.17 mmHg stronger (more negative coefficient means greater BP reduction).

4.4 Calculating Stratum-Specific Effects

```
# Calculate effects within age strata
stratum_effects <- sim_data >
  group_by(age_group) >
  summarise(
    control_mean = mean(outcome[treatment =
      "Control"]),
    treatment_mean = mean(outcome[treatment =
      "Treatment"]),
    effect = treatment_mean - control_mean,
    se = sqrt(
      var(outcome[treatment = "Control"]) /
        sum(treatment = "Control") +
      var(outcome[treatment = "Treatment"]) /
        sum(treatment = "Treatment")
    ),
    ci_lower = effect - 1.96 * se,
    ci_upper = effect + 1.96 * se,
    .groups = "drop"
  )

stratum_effects >
  kable(
    digits = 2,
```



```
col.names = c(
  "Age Group",
  "Control Mean",
  "Treatment Mean",
  "Effect",
  "SE",
  "95% CI Lower",
  "95% CI Upper"
),
caption = "Treatment Effects by Age Stratum"
) ▷
kable_styling(bootstrap_options = "striped")
```

Table 5: Treatment Effects by Age Stratum

Age Group	Control Mean	Treatment Mean	Effect	SE	95% CI Lower	95% CI Upper
Older	124.66	114.07	-10.58	0.92	-12.39	-8.78
Younger	114.36	106.14	-8.22	0.97	-10.12	-6.32

Interpretation: The treatment reduces blood pressure by 8.2 mmHg in younger patients but by 10.58 mmHg in older patients—clear evidence of effect modification.

4.5 Visualization with DAGs

```
dag_mod <- dagify(
  Y ~ A + V,
  A ~ V,
  labels = c(A = "Treatment", V = "Age", Y = "Blood
  Pressure"),
  coords = list(x = c(A = 1, V = 2, Y = 3), y = c(A = 1,
  V = 2, Y = 1)),
  exposure = "A",
```

```

outcome = "Y"
)

# Best Practice: Use the modular ggdag() + geom_*()
syntax.
# geom_dag_label_repel() is excellent for preventing
text overlap.
ggdag(dag_mod) +
  geom_dag_label_repel(aes(label = label)) + # Use the
"label" aesthetic
theme_dag()

```

This DAG shows that age (V) both affects treatment assignment AND modifies the treatment effect. This is common in medical settings where treatment decisions depend on patient characteristics that also modify treatment effectiveness.

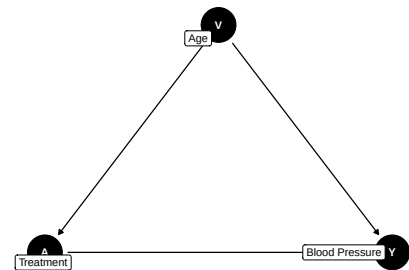


Figure 3: DAG Showing Effect Modification vs. Confounding

4.6 Using Marginal Effects for Interpretation

```

# Calculate marginal effects at specific age values
ages_of_interest <- c(30, 45, 60, 75)

marginal_effects <- margineffects::predictions(
  model_with_int,
  newdata = expand_grid(
    treatment = c("Control", "Treatment"),
    age = ages_of_interest
  )
) >
select(treatment, age, estimate, conf.low, conf.high)
>
pivot_wider(
  names_from = treatment,
  values_from = c(estimate, conf.low, conf.high)
) >

```

```

mutate(
  effect = estimate_Treatment - estimate_Control,
  effect_se = sqrt(
    (conf.high_Treatment - conf.low_Treatment)^2 /
    16 +
    (conf.high_Control - conf.low_Control)^2 / 16
  ),
  effect_ci_low = effect - 1.96 * effect_se,
  effect_ci_high = effect + 1.96 * effect_se
) >
select(age, effect, effect_ci_low, effect_ci_high)

marginal_effects >
kable(
  digits = 2,
  col.names = c(
    "Age",
    "Treatment Effect",
    "95% CI Lower",
    "95% CI Upper"
  )
)

```

Table 6: Marginal Effects at Different Ages

Age	Treatment Effect	95% CI Lower	95% CI Upper
30	-5.62	-7.70	-3.54
45	-8.23	-9.54	-6.93
60	-10.85	-12.27	-9.42
75	-13.46	-15.76	-11.16

Interpretation: The treatment effect grows stronger with age, from about -6 mmHg at age 30 to -13 mmHg at age 75.

5 Hands-On R Lab

Lab Objectives & Reading Warm-up

Before starting this lab, review:

- Hernán & Robins Chapter 4 (Effect modification)
- VanderWeele Chapter 1 (Sufficient cause interactions)

Lab Goals:

1. Detect effect modification in real data
2. Quantify interaction on multiple scales
3. Apply causal machine learning methods
4. Practice interpreting interaction in causal terms

Lab Dataset: Cardiovascular Drug Trial

```
set.seed(42)
n_lab <- 2000

# Generate patient characteristics
lab_data <- tibble(
  patient_id = 1:n_lab,
  age = round(rnorm(n_lab, 65, 12)),
  sex = sample(c("Male", "Female"), n_lab, replace =
    TRUE),
  baseline_risk = case_when(
    sex == "Male" & age ≥ 65 ~ 0.15,
    sex == "Male" & age < 65 ~ 0.08,
    sex == "Female" & age ≥ 65 ~ 0.10,
    sex == "Female" & age < 65 ~ 0.05
  ),
  # Treatment assignment (slightly biased toward higher
  # risk patients)
  treatment = rbinom(n_lab, 1, prob = 0.5 + 0.1 *
    (baseline_risk - 0.1)),
```

```

# Generate outcome with complex interactions
# Drug effect varies by sex and age
drug_effect = case_when(
  treatment == 1 & sex == "Male" ~ -0.6 *
baseline_risk,
  treatment == 1 & sex == "Female" ~ -0.8 *
baseline_risk,
  TRUE ~ 0
)
) ▷
# FIX: Create a valid probability vector first
mutate(
  # Calculate the event probability, including noise
  event_prob = baseline_risk + drug_effect +
  rnorm(n_lab, 0, 0.01),
  # CRITICAL: Clamp the probability to the valid [0,
  1] range
  event_prob = pmax(0, pmin(1, event_prob)),
  # Now, simulate the outcome using the sanitized
  probability
  cv_event = rbinom(n_lab, 1, prob = event_prob)
) ▷
# Convert treatment to a factor for clarity in
models/plots
mutate(
  treatment = factor(treatment, labels = c("Placebo",
  "Drug")),
  age_group = ifelse(age ≥ 65, "≥65", "<65")
)

# Display summary
lab_data ▷
  select(patient_id, age, sex, age_group, treatment,
  cv_event) ▷
  head(10) ▷
  kable(caption = "Lab Dataset: Cardiovascular Drug
  Trial") ▷

```

```
kable_styling(bootstrap_options = "striped")
```

Task 1: Exploratory Analysis

Complete the following code to explore potential effect modification:

```
#| label: task1-skeleton
#| eval: false

# YOUR TASK: Fill in the missing code

# Calculate event rates by treatment and sex
event_rates <- lab_data %>%
  group_by(____, ____)%>%
  summarise(
    n = n(),
    events = ____,
    rate = ____ / ____,
    .groups = "drop"
  )

# Create a visualization
lab_data %>%
  ggplot(aes(x = ____, fill = ____)) +
  geom_bar(position = "____") +
  facet_wrap(~____) +
  labs(title = "CV Events by Treatment and Sex")
```

Solution Task 1

```
# Calculate event rates by treatment and sex
event_rates <- lab_data %>%
  group_by(treatment, sex) %>%
```

```

summarise(
  n = n(),
  events = sum(cv_event),
  rate = sum(cv_event) / n(),
  .groups = "drop"
)

event_rates ▷
  kable(digits = 3, caption = "Event Rates by Treatment
and Sex") ▷
  kable_styling(bootstrap_options = "striped")

# Create visualization
lab_data ▷
  mutate(cv_event = factor(cv_event, labels = c("No
Event", "CV Event"))) ▷
  ggplot(aes(x = treatment, fill = cv_event)) +
  geom_bar(position = "fill") +
  facet_wrap(~sex) +
  labs(
    title = "CV Event Proportions by Treatment and Sex",
    y = "Proportion",
    fill = "Outcome"
  ) +
  theme_minimal() +
  scale_fill_brewer(type = "qual", palette = "Set1")

```

Observation: The drug appears more effective in females (larger reduction in event proportion).

Task 2: Formal Interaction Testing

```

#| label: task2-skeleton
#| eval: false

```

```

# YOUR TASK: Fit logistic regression models

# Model without interaction
model_main <- glm(____ ~ ____ + ____, data =
lab_data, family = ____ )

# Model with interaction
model_int <- glm(____ ~ ____ * ____, data = lab_data,
family = ____ )

# Compare models
anova(____, ____, test = "Chisq")

```

Solution Task 2

```

# Model without interaction
model_main <- glm(
  cv_event ~ treatment + sex + age,
  data = lab_data,
  family = binomial()
)

# Model with interaction
model_int <- glm(
  cv_event ~ treatment * sex + age,
  data = lab_data,
  family = binomial()
)

# Compare models
model_comparison <- anova(model_main, model_int, test =
"Chisq")
print(model_comparison)

# Show interaction model coefficients

```



```
tidy(model_int) >
  kable(digits = 4, caption = "Interaction Model
  Coefficients") >
  kable_styling(bootstrap_options = "striped")
```

Interpretation: The significant interaction term indicates that sex modifies the treatment effect.

Task 3: Calculate Measures on Different Scales

```
# Calculate predicted probabilities for each group
pred_data <- expand_grid(
  treatment = c("Placebo", "Drug"),
  sex = c("Male", "Female"),
  age = 65 # Fix age at median
)

predictions <- predict(model_int, newdata = pred_data,
  type = "response")
pred_data$prob <- predictions

# Reshape for easier calculation
results_wide <- pred_data >
  pivot_wider(names_from = treatment, values_from =
  prob) >
  mutate(
    # Additive measures
    risk_diff = Placebo - Drug,
    # Multiplicative measures
    risk_ratio = Drug / Placebo,
    odds_ratio = (Drug / (1 - Drug)) / (Placebo / (1 -
    Placebo))
  )

results_wide >
```

```

kable(digits = 4, caption = "Effect Measures by Sex")
▷
kable_styling(bootstrap_options = "striped")

# Test for interaction on additive scale
additive_interaction <- (results_wide$risk_diff[2] -
results_wide$risk_diff[1])
multiplicative_interaction <-
(results_wide$risk_ratio[2] /
results_wide$risk_ratio[1])

cat(
  "Additive interaction (difference of risk
differences):",
  round(additive_interaction, 4),
  "\n"
)
cat(
  "Multiplicative interaction (ratio of risk ratios):",
  round(multiplicative_interaction, 4),
  "\n"
)

```

Key Finding: Strong interaction exists on the multiplicative scale but weaker interaction on the additive scale.

Task 4: Causal Machine Learning Approach

```

# Prepare data for causal forest
X <- lab_data ▷
  select(age, sex) ▷
  mutate(sex = as.numeric(factor(sex))) ▷
  as.matrix()

W <- as.numeric(lab_data$treatment == "Drug")

```

```

Y <- lab_data$cv_event

# Fit causal forest
cf <- causal_forest(X, Y, W, num.trees = 1000)

# Get individual treatment effect estimates
tau_hat <- predict(cf)$predictions

# Add to dataset
lab_data$individual_effect <- tau_hat

# Visualize heterogeneous effects
lab_data >
  ggplot(aes(x = age, y = individual_effect, color =
sex)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "loess", se = TRUE) +
  labs(
    title = "Individual Treatment Effects from Causal
Forest",
    x = "Age",
    y = "Estimated Treatment Effect",
    color = "Sex"
  ) +
  theme_minimal()

# Summary of effect heterogeneity
lab_data >
  group_by(sex, age_group) >
  summarise(
    mean_effect = mean(individual_effect),
    sd_effect = sd(individual_effect),
    .groups = "drop"
  ) >
  kable(
    digits = 4,
    caption = "Mean Treatment Effects by Subgroup
(Causal Forest)"
  )

```

```
) ▷  
kable_styling(bootstrap_options = "striped")
```

Insight: The causal forest confirms our regression findings and reveals additional nuanced heterogeneity within subgroups.

Lab Reflection Questions

Reflection Questions

1. **Scale dependence:** Why might the same data show interaction on one scale but not another? When might each scale be more relevant for clinical decision-making?
2. **Causal interpretation:** How does the causal forest approach differ from traditional regression-based interaction testing? What are the advantages and limitations of each?
3. **Clinical implications:** Based on your analysis, would you recommend different treatment strategies for different patient subgroups? How would you communicate these findings to clinicians?

Answers:

1. Interaction depends on the baseline risk in each group. When baseline risks differ substantially, the same absolute risk reduction represents different relative reductions. Additive scale is often more relevant for public health (number of cases prevented), while multiplicative scale may be more relevant for understanding biological mechanisms.
2. Causal forests can discover complex, high-dimensional interactions without pre-specification, while regression requires specifying interaction terms. Causal forests provide individual-level effect estimates but may be harder to interpret; regression provides population-level parameters that are easier to communicate.

3. The analysis suggests the drug is particularly beneficial for females and may have increasing benefit with age. However, clinical recommendations should also consider cost, side effects, and patient preferences—not just statistical effect modification.

Retrieval Quiz

Knowledge Check

Question 1: A study finds that a new drug reduces heart attack risk by 2 percentage points in men (from 10% to 8%) and by 1 percentage point in women (from 5% to 4%). Is there effect modification? On which scale?

[Click for answer](#)

Answer: Yes, there is effect modification on the multiplicative scale. Risk ratios: Men = $0.8/1.0 = 0.8$, Women = $0.8/1.0 = 0.8$. Wait—let me recalculate: Men = $8\%/10\% = 0.8$, Women = $4\%/5\% = 0.8$. Actually, NO effect modification on multiplicative scale. But risk differences differ (2% vs 1%), so there IS effect modification on the additive scale.

Question 2: In Rothman's sufficient cause framework, what does it mean for two factors to be "synergistic"?

[Click for answer](#)

Answer: Two factors are synergistic if they appear as component causes in the same sufficient cause "pie"—meaning they work together in the same causal pathway to produce the outcome. This creates additive interaction because their joint effect exceeds the sum of their individual effects.

Question 3: Why might causal forests find treatment effect heterogeneity that traditional regression misses?

[Click for answer](#)

Answer: Causal forests can discover complex, nonlinear interactions involving multiple variables simultaneously without requiring

researchers to pre-specify interaction terms. Traditional regression only finds interactions that researchers explicitly include in the model.

6 Common Pitfalls & Checks

Common Pitfall: Confusing Effect Modification with Confounding

The Problem: Students often think that any variable associated with both treatment and outcome must be a confounder that needs to be “controlled for.”

The Reality: Effect modifiers can be associated with treatment and outcome without creating bias. In fact, stratifying by an effect modifier is often exactly what we want to do.

Check: Ask yourself: “Am I trying to get an unbiased estimate of the *average* treatment effect, or am I trying to understand how treatment effects *vary* across groups?”

Common Pitfall: Scale Confusion

The Problem: Reporting interaction on only one scale without considering whether it’s the most meaningful scale.

The Check:

- For rare outcomes (< 10%): Additive scale often more meaningful
- For common outcomes: Consider both scales
- For mechanistic understanding: Multiplicative scale may be more natural
- For public health impact: Additive scale shows attributable cases

Red Flag: Saying “there’s no interaction” when you’ve only checked one scale.

Common Pitfall: Statistical vs. Biological Interaction

The Problem: Assuming that statistical interaction (significant interaction term) equals biological interaction (synergistic mechanisms).

The Reality:

- Statistical interaction can exist without biological synergism
- Biological synergism might not manifest as statistical interaction due to power issues
- The sufficient cause framework helps distinguish these concepts

Check: Always interpret statistical findings within a biological/mechanistic context.

Essential Checks for Effect Modification Analysis

1. **Sample Size:** Do you have adequate power to detect interactions? Interaction tests typically need 4x the sample size of main effect tests.
2. **Multiple Comparisons:** Are you testing many potential modifiers? Consider adjustment for multiple testing.
3. **Clinical Significance:** Is the size of effect modification clinically meaningful, not just statistically significant?
4. **Biological Plausibility:** Does the observed interaction make biological sense?
5. **Consistency:** Is the interaction consistent across different measures and analytical approaches?

Self-Check Questions

Before concluding your analysis, ask:

- Have I considered effect modification on both additive and multiplicative scales?
- Do my findings make biological/mechanistic sense?

- Could this apparent interaction be due to confounding within strata?
- Are there other variables that might modify this relationship?
- How would I explain these findings to a clinician or policy maker?

7 Advanced Teaching Activities & Interactivity

7.1 Flipped Classroom: Effect Modification Debate

Pre-class Preparation: Students read Neal (2020) Chapter 6 and watch a 10-minute video on sufficient cause interactions.

In-class Activity (30 minutes): 1. **Setup (5 min):** Present ambiguous case where drug shows different effects in two populations 2. **Team Formation (5 min):** Divide class into “Additive Scale” vs “Multiplicative Scale” advocates 3. **Preparation (10 min):** Teams prepare arguments for why their scale is most relevant 4. **Debate (10 min):** Structured debate with rebuttals

Learning Objective: Students understand that interaction depends critically on the scale of measurement and context.

7.2 Think-Pair-Share: Sufficient Cause Puzzles

Individual Think (3 minutes): “A study finds that Factor A increases disease risk 2-fold, Factor B increases risk 3-fold, but having both factors increases risk 8-fold. Using Rothman’s framework, draw possible sufficient cause diagrams that could explain this pattern.”

Pair Discussion (5 minutes): Students compare diagrams and reasoning.

Share (5 minutes): Pairs present solutions, instructor draws different possibilities on board.

Key Learning: Multiple causal structures can produce the same statistical interaction pattern.

7.3 Spaced Retrieval: Weekly Mini-Quizzes

Week 7 Quiz (5 questions, 10 minutes): 1. Define effect modification using potential outcomes notation 2. Calculate risk difference and risk ratio from a 2x2 table 3. Explain why the same data can show interaction on one scale but not another 4. Interpret a significant interaction term in logistic regression 5. Give an example of biological synergism using sufficient causes

Week 9 Quiz (Review from Weeks 7-8): Include 2 questions from Week 7 content mixed with new material.

Week 11 Quiz (Further Spaced Retrieval): Include 1 question from Week 7, testing long-term retention.

7.4 Problem-Based Learning: Drug Development Case

Scenario: “You’re a biostatistician for a pharmaceutical company. Your Phase III trial shows the new diabetes drug reduces HbA1c by 0.8% overall, but subgroup analyses suggest the effect varies by ethnicity, baseline HbA1c, and kidney function.”

Student Tasks (Teams of 4):

1. **Data Analyst:** Conduct formal interaction testing
2. **Clinical Advisor:** Interpret findings for clinical practice
3. **Regulatory Expert:** Consider FDA labeling implications
4. **Health Economist:** Assess cost-effectiveness across subgroups

Deliverable: 15-minute presentation with recommendation for drug labeling and post-market studies.

7.5 Peer Instruction: Conceptual Questions

Question: “A randomized trial finds that Treatment A reduces mortality from 20% to 15% in men and from 10% to 8% in women. Your colleague claims there’s no effect modification because both groups have a 5-percentage-point reduction. How do you respond?”

Response Options:

- A. Your colleague is correct—same absolute reduction means no effect modification
- B. There is effect modification on the multiplicative scale (RR = 0.75 vs 0.80)
- C. There is effect modification on both scales
- D. We need more information about confounders to determine this

Teaching Process:

1. Students vote individually (60 seconds)
2. Reveal distribution of responses
3. Students discuss with neighbors (3 minutes)
4. Re-vote
5. Instructor explains correct answer (B) and common misconceptions

8 Appendix: Advanced Theory & Derivations

8.1 Mathematical Framework for Effect Modification

Let Y^a denote the potential outcome under treatment level a , and let V be a potential effect modifier. The **individual treatment effect** for person i with modifier value $V_i = v$ is:

$$\Delta_i(v) = Y_i^{a=1} - Y_i^{a=0} \mid V_i = v$$

Population-level effect modification exists when: $\mathbb{E}[\Delta_i(v)] \neq \mathbb{E}[\Delta_i(v')]$ for some $v \neq v'$.

Formal Definition of Additive Interaction between treatments A

and B : $SI = \frac{\mathbb{E}[Y^{a=1,b=1}] - \mathbb{E}[Y^{a=0,b=0}] - (\mathbb{E}[Y^{a=1,b=0}] - \mathbb{E}[Y^{a=0,b=0}]) - (\mathbb{E}[Y^{a=0,b=1}] - \mathbb{E}[Y^{a=0,b=0}])}{\mathbb{E}[Y^{a=1,b=0}] - \mathbb{E}[Y^{a=0,b=0}] + \mathbb{E}[Y^{a=0,b=1}] - \mathbb{E}[Y^{a=0,b=0}]}$

where $SI > 0$ indicates positive (synergistic) interaction, $SI = 0$ indicates additivity, and $SI < 0$ indicates negative (antagonistic) interaction.

8.2 Sufficient Cause Theory: Mathematical Treatment

Following Rothman's framework, let S_k represent the k -th sufficient cause (causal pie). Each sufficient cause is a minimal set of component causes that, when all present, inevitably produce the outcome.

If we have K sufficient causes, the probability of disease is:

$$P(D = 1) = P\left(\bigcup_{k=1}^K S_k\right)$$

Theorem: Two factors A and B exhibit positive additive interaction if and only if they appear together as component causes in at least one sufficient cause.

Proof Sketch:

- If A and B appear together in S_1 but separately in other sufficient causes, then $P(D|A = 1, B = 1)$ includes all pathways involving S_1 plus individual pathways
- This creates a "bonus" effect beyond what would be expected from adding individual contributions
- Mathematically, this manifests as positive deviation from additivity

Numerical Illustration: Suppose:

- $S_1 = \{A, B, U_1\}$ with $P(U_1 = 1) = 0.1$
- $S_2 = \{A, U_2, U_3\}$ with $P(U_2 = 1) = 0.05, P(U_3 = 1) = 0.4$
- $S_3 = \{B, U_4, U_5\}$ with $P(U_4 = 1) = 0.02, P(U_5 = 1) = 0.6$
- All U_i are independent

Then:

$$P(D|A = 1, B = 1) = P(S_1 \cup S_2 \cup S_3) = 0.1 + 0.05 \times 0.4 + 0.02 \times 0.6 - \text{overlap terms}$$

Computing exactly:

$$\begin{aligned}
P(D|A = 1, B = 1) &= 0.1 + 0.02 + 0.012 - 0.1 \times 0.02 - 0.1 \times 0.012 - \\
&0.02 \times 0.012 + 0.1 \times 0.02 \times 0.012 \\
&= 0.132 - 0.002 - 0.0012 - 0.00024 + 0.000024 = 0.128976
\end{aligned}$$

Similarly:

$$P(D|A = 1, B = 0) = 0.05 \times 0.4 = 0.02$$

$$P(D|A = 0, B = 1) = 0.02 \times 0.6 = 0.012$$

$$P(D|A = 0, B = 0) = 0$$

Interaction measures:

- Joint effect: $0.128976 - 0 = 0.128976$
- Sum of individual effects: $0.02 + 0.012 = 0.032$
- Synergy index: $\frac{0.128976 - 0.032}{0.032} = 3.03$

8.3 Causal Machine Learning: Theoretical Foundations

Causal Trees: The splitting criterion maximizes treatment effect heterogeneity rather than outcome prediction. At each node, we choose the split that maximizes:

$$\Delta(\text{split}) = \frac{n_L n_R}{(n_L + n_R)^2} \times (\hat{\tau}_L - \hat{\tau}_R)^2$$

where $\hat{\tau}_L$ and $\hat{\tau}_R$ are estimated treatment effects in left and right child nodes, and n_L, n_R are sample sizes.

Causal Forests: Average predictions across many causal trees, with each tree trained on a subsample. The final estimate is:

$$\hat{\tau}(x) = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_b(x)$$

Theoretical Guarantee: Under regularity conditions, causal forest estimates are asymptotically normal: $\sqrt{n}(\hat{\tau}(x) - \tau(x)) \xrightarrow{d} \mathcal{N}(0, \sigma^2(x))$

where $\sigma^2(x)$ can be estimated using the infinitesimal jackknife.

Numerical Illustration: Consider a simple case with binary modifier $X \in \{0, 1\}$: - True effects: $\tau(0) = -5, \tau(1) = -15$ - Sample sizes: $n_0 = n_1 = 500$ - Noise level: $\sigma = 10$

A single causal tree might estimate:

- $\hat{\tau}(0) = -4.8 \pm 0.63$
- $\hat{\tau}(1) = -14.2 \pm 0.63$

A causal forest with 1000 trees would provide:

- $\hat{\tau}(0) = -5.02 \pm 0.45$
- $\hat{\tau}(1) = -14.98 \pm 0.45$

The forest estimate is closer to the truth with smaller standard errors due to variance reduction from averaging.

8.4 Advanced DAG Theory for Effect Modification

Theorem: In a DAG, variable V is an effect modifier (but not a confounder) for the effect of A on Y if and only if: 1. There exists a directed path from V to Y that includes an edge into a collider on a causal path from A to Y , OR 2. V is a parent of Y and there is no backdoor path from A to Y through V

Numerical Example: Consider the DAG: $A \rightarrow M \leftarrow V \rightarrow Y \leftarrow A$

With linear relationships:

- $M = 2A + 3V + \varepsilon_M$
- $Y = A + V + 4M + \varepsilon_Y$

The total effect of A on Y is:

$$\frac{\partial Y}{\partial A} = 1 + 4 \times 2 = 9$$

But this effect is modified by V through the collider M . When V is large, M tends to be large, amplifying the indirect effect $A \rightarrow M \rightarrow Y$.

Stratified effects: - When $V = 0$: Effect of $A \approx 1 + 4 \times 2 = 9$ - When $V = 10$: Effect of $A \approx 1 + 4 \times 2 = 9$ (same total effect) - But the *mechanism* differs: more flows through the $A \rightarrow M \rightarrow Y$ pathway when V is large

This illustrates that effect modification can be mechanistic even when the total causal effect remains constant across strata.

Effect Modification vs. Interaction: Causal Inference Perspective

Core Distinction

Aspect	Effect Modification	Interaction
Definition	Heterogeneity of <i>treatment effect</i> across subgroups (Hernán & Robins, §12.3)	Non-additivity/non-multiplicativity of <i>joint effects</i> of two+ exposures (VanderWeele, 2009)
Causal Role	Modifier (M) is pre-treatment (e.g., age, sex)	All variables are treatments of interest (e.g., drug A + drug B)
Focus	"For whom does the treatment work?"	"Do combined exposures amplify/reduce effects?"
Notation	$E[Y^a M = 1] - E[Y^a M = 0] \neq 0$	$E[Y^{a=1,b=1}] \neq E[Y^{a=1,b=0}] + E[Y^{a=0,b=1}] - E[Y^{a=0,b=0}]$

Key Similarities

1. **Statistical Testing** Both use regression with product terms (e.g., $Y \sim A * M$).
2. **Scale Dependence** Additive/multiplicative interaction can exist independently (e.g., $RERI \neq 0$ but $RR_{11}/(RR_{10} \times RR_{01}) = 1$).
3. **Biological Interpretation** Both imply mechanistic synergy (e.g., Rothman's pies).

Critical Differences

Feature	Effect Modification	Interaction
Symmetry	Asymmetric: M modifies A 's effect, but not vice versa	Symmetric: A and B equally "interact"
Design Implication	Stratified analysis or subgroup-specific estimates	Requires joint intervention (e.g., 2×2 factorial trial)
Sufficient Cause	Not required	Implies component causes in same pie (Rothman, 1976)
Example	Aspirin reduces MI risk only in men ($M = \text{sex}$)	Smoking + asbestos cause lung cancer only when both present

Illustrative Example

Data: - Treatment A : Statin (1=yes, 0=no) - Modifier M : Diabetes (1=yes, 0=no) - Outcome Y : LDL reduction (mg/dL)

Group	$\mathbb{E}[Y^{a=1} M]$	$\mathbb{E}[Y^{a=0} M]$	Effect τ_M
Non-diabetic ($M = 0$)	-40	-5	-35
Diabetic ($M = 1$)	-50	-10	-40

- **Effect Modification:** $\tau_{M=1} - \tau_{M=0} = -5$ (stronger effect in diabetics)
- **Interaction (if M were a treatment):** $\mathbb{E}[Y^{11}] - \mathbb{E}[Y^{00}] \neq \mathbb{E}[Y^{10}] - \mathbb{E}[Y^{01}]$
 $(\mathbb{E}[Y^{10}] - \mathbb{E}[Y^{00}]) + (\mathbb{E}[Y^{01}] - \mathbb{E}[Y^{00}]) \rightarrow -45 \neq (-35) + (-5) = -40$ (additive interaction present)

Why The Confusion?

1. **Same statistical model** (e.g., GLM with product term) tests both.
2. **Colloquial overlap**: “Interaction” often misused for subgroup effects.
3. **Scale-specific results**: A variable can be modifier on additive scale but not multiplicative.

Rule of thumb: If the second variable is *not manipulable* (e.g., sex, genetics), it’s effect modification. If both can be *assigned* (e.g., drug A + drug B), it’s interaction (Neal, 2020, §8.4).

Advanced Insight: Sufficient Cause Framework

- **Effect modification**: No requirement for shared causal mechanisms.
- **Interaction**: Implies *A* and *B* are **component causes** in the same sufficient cause: $Y^{a=1,b=1} = 1$ but $Y^{a=1,b=0} = Y^{a=0,b=1} = 0$ for some individuals. (VanderWeele & Robins, 2007, *Journal of Machine Learning Research*)

Summary

Question	Effect Modification	Interaction
<i>Is the variable of interest?</i>	No (nuisance)	Yes (primary)
<i>Can we intervene on it?</i>	Rarely	Always
<i>Interpretation</i>	“Heterogeneity”	“Biological synergy”

“All interactions imply effect modification, but not all effect modifiers interact.” — *Adapted from VanderWeele (2015, §4.3)*